

Calculus II

Integrals of the form $\int \sqrt{ax^2 + c} dx, a, c > 0$

Todor Milev

2019

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$$\int \sqrt{x^2 + 1} \, dx =$$

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 &= \int \frac{1}{\textcolor{red}{27} \cot^2 \theta \sqrt{\textcolor{red}{?}}} \left(\textcolor{red}{?} \right) d\theta
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$$x = 3 \cot \theta$$

$$\theta \in (0, \pi)$$

Example

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
 &= \int \frac{1}{(3 \cot \theta)^2 3 \sqrt{\cot^2 \theta + 1}} d(3 \cot \theta) \\
 &= \int \frac{1}{27 \cot^2 \theta \sqrt{\csc^2 \theta}} \left(? \right) d\theta
 \end{aligned}$$

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$$\theta \in (0, \pi) \Rightarrow$$

$$\csc \theta > 0$$

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 \end{aligned}$$

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 &= \frac{1}{9} \int \frac{du}{u^2}
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 $x = 3 \cot \theta$
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Set $u = \cos \theta$

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 &= \frac{1}{9} \int \frac{du}{u^2} = -\frac{1}{9u} + C = -\frac{\sec \theta}{9} + C \\
 &= -\frac{\sqrt{x^2 + 9}}{9x} + C
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